UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages

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PART A (80 MARKS)

QUESTION 1

a) State TWO (2) reasons why software is insecure according to the lecture.

(2 marks)

b) Refer to your answer in (a). Explain six ways Windows security mechanisms manage users, objects and access, and why their complexity can itself become a security risk.

(6 marks)

QUESTION 2

a) Explain how a substitution cipher works and what serves as its key.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.

UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

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QUESTION 3

a) State TWO (2) formal security models that address integrity or conflicts of interest.

(2 marks)

b) Describe six (6) points about the access control matrix, access control list and capability structures, covering how each is organised and its stated advantages or disadvantages.

(6 marks)

QUESTION 4

a) List TWO (2) goals of access control and state what a reference monitor does.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) Explain two characteristics of Trojan horse malware that distinguish its behaviour on an infected host.

(2 marks)

b) Describe six techniques by which malware infects a host or propagates itself, and the effect of each technique.

(6 marks)

QUESTION 6

a) State the two stages of the biometric authentication process.

(2 marks)

b) State six facts about biometric mechanisms and technologies as presented in the course, covering the definition, the categories, and specific biometric technologies.

(6 marks)

QUESTION 7

a) Explain where NIDS sensors should be placed and why such placement matters.

(2 marks)

b) State six preventive network security mechanisms and, for each, one function it performs in protecting an organization's network.

(6 marks)

QUESTION 8

a) Explain two password security controls used to protect password files and accounts.

(2 marks)

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b) Discuss three design criteria of Kerberos and three motivations for adopting single sign-on in distributed environments.

(6 marks)

QUESTION 9

a) Explain how a chain of trust is established in trusted computing using the TPM.

(2 marks)

b) Explain six components and functions of the NGSCB architecture, focusing on the Nexus and the Nexus Computing Agent.

(6 marks)

QUESTION 10

a) List two impacts of threats on confidentiality, integrity or availability.

(2 marks)

b) Explain how the Samy worm used cross-site scripting to infect MySpace user profiles, and what made the attack so widespread.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Explain how an intrusion detection system contributes to detecting and defending against attacks, and outline the limitations of misuse-based detection.

(5 marks)

b) Explain the distinction between a vulnerability, a threat and an attack, and describe how organisations counter threats arising from forces of nature and from malicious web input.

(5 marks)

QUESTION 2

a) State the meaning of morals, ethics and laws, and state two limitations of relying on a professional code of ethics.

(5 marks)

b) Discuss what a professional code of ethics and conduct for computing staff should contain and how it should guide their behaviour, with reference to recognised codes and the ethical issues they address.

(5 marks)

END OF QUESTION PAPER

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